AD 502 – Machine Learning

Pre-Reading Material – Lecture 32

Neural Network-III

Types and Applications of Neural Networks

Neural networks are available in different architectures designed to process different types of data. The selection of an appropriate architecture depends on the problem and the nature of the input data.

Major Types of Neural Networks

1. Feedforward Neural Network (FNN)

Information moves from the input layer through hidden layers to the output layer without forming cycles. These networks are commonly used for classification and regression.

2. Convolutional Neural Network (CNN)

CNNs are particularly suitable for images and visual data. They learn features such as edges, shapes and objects. Applications include facial recognition and medical image analysis.

3. Recurrent Neural Network (RNN)

RNNs are designed to process sequential data, using information from earlier steps to help process later steps. They have been used in speech recognition and time-series analysis.

4. Long Short-Term Memory (LSTM)

LSTM is a type of recurrent neural network designed to learn longer-term dependencies in sequential data. It can be applied to electricity-demand forecasting and other time-series problems.

5. Transformer Network

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Transformers use attention mechanisms to identify relationships between elements in data. They are widely used in language processing, translation, text summarization and other AI applications.

Real-World Applications

Neural networks are used in several domains:

- Healthcare: Medical image analysis and decision support.
- Banking: Fraud detection and risk assessment.
- Education: Student-performance prediction.
- Energy: Electricity-demand forecasting and renewable-energy prediction.
- Transportation: Object detection and driver-assistance systems.
- Natural Language Processing: Text classification, translation and conversational systems.
- Manufacturing: Fault detection and predictive maintenance.

Challenges of Neural Networks

Despite their advantages, neural networks face several challenges:

- They may require large and high-quality datasets.
- Training complex networks can be computationally expensive.
- Their predictions may be difficult to interpret.
- Biased training data can lead to unfair outcomes.
- Sensitive data must be protected during training and deployment.

Selecting an Appropriate Neural Network

The type of data provides a useful starting point for model selection. CNNs are often used for image processing, RNNs and LSTMs for sequential data, and transformers for complex language tasks. Feedforward networks are useful for many general prediction problems.

Conclusion: Different neural network architectures address different learning tasks. Appropriate model selection, data quality and performance evaluation are essential for developing reliable neural network applications.

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